

KAMALESH G

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GitHub LinkedIn Portfolio

Summary

Computer Science student skilled in full-stack MERN software development, REST APIs, and modern AI solutions using Python and LangChain. Strong foundation in Data Structures and Algorithms, Database Management Systems, and Object-Oriented Programming, with strong analytical and problem solving abilities. Adept at transforming complex requirements into practical, scalable systems and eager to contribute to a dynamic engineering team.

Education

Bachelor of Engineering in Computer Science and Engineering 2023 – 2027
PSG College of Technology, Coimbatore

Coursework: Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Data Structures and Algorithms, Object-Oriented Programming

Class XII & X 2023, 2021
Sri Gurukulam Senior Secondary School, Hosur

Skills

Programming Languages: Java, Python, JavaScript

Web Development: React.js, Node.js, HTML, CSS

Databases & Tools: MongoDB, Qdrant (Vector DB), MySQL, Git, GitHub

AI & Machine Learning: Scikit-learn, Pandas, NumPy, RAG Pipelines

Projects

Research Paper Summarizer & Analyzer | GitHub

- Developed a Retrieval-Augmented Generation (RAG) application using FastAPI and Qdrant to ingest, chunk, and perform semantic search across multi-document research papers.
- Engineered a multi-agent system using LangChain, implementing an intelligent routing layer to direct queries to specialized Summarizer, Analyzer, and Chatbot agents based on user intent.
- Engineered pipeline logging to track system performance, consistently monitoring vector retrieval latencies under 3.7 seconds and isolating LLM generation metrics across distinct agent workflows.

Community Connect | GitHub

- Developed a full-stack MERN platform that connects NGOs with volunteers through a requirement-driven ecosystem for donations, communication, and community engagement.
- Implemented JWT-based authentication, role-specific workflows for NGOs, volunteers, and administrators, and an NGO verification system for secure platform management.
- Integrated real-time requirement feeds, activity tracking, chat functionality, notifications, and Razorpay payment processing to streamline volunteer participation and donations.

LangChain Learning Assistant | GitHub

- Built an intelligent multi-agent AI learning assistant using LangChain, OpenRouter, and Streamlit to orchestrate automated topic explanations, quiz generation, and performance evaluation.
- Designed an architecture featuring specialized Planner, Explainer, Quiz, Evaluator, and Router agents to execute structured, state-driven educational workflows and dynamic task routing.
- Implemented robust session state management, structured JSON data validation using Pydantic, middleware logging, and custom interactive learning workflows.

Certifications & Activities

Certifications: LangChain Foundations | Building RAG Applications Using MongoDB | CRUD Operations in MongoDB | Python Essentials 2

- Head, PSG RadioHub – Led content creation and event coordination.
- Member, IEEE Student Branch, PSG College of Technology.